

SingleRAN

Access Control based on 802.1x Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue does not introduce any changes to SRAN21.1 01 (2025-03-10).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and New Radio (NR).

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

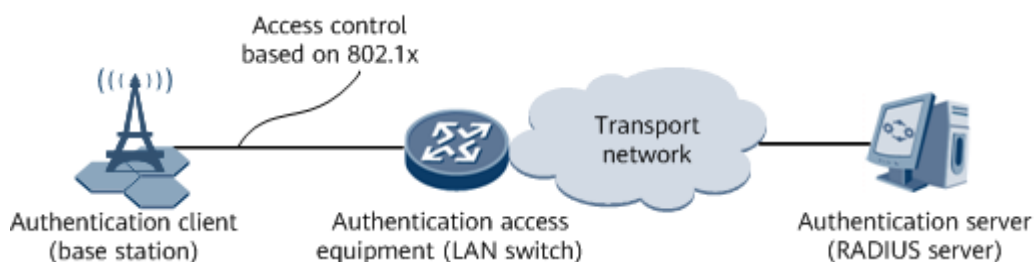
RA T	Feature ID	Feature Name	Chapter/Section
LTE FD D	LOFD-003015	Access Control based on 802.1x	4 Access Control based on 802.1x
NB- IoT	MLOFD-003015	Access Control based on 802.1x	
LTE TD D	TDLOFD-003015	Access Control based on 802.1x	
NR	FBFD-010023	Security Mechanism (Access Control Based on 802.1x)	

3 Overview

IEEE 802.1x, complying with IEEE 802 standards for local area network (LAN) access control, is an IEEE standard for port-based network access control. Port-based network access control requires that the authentication access equipment in the LAN perform identity authentication and access control on users or devices that visit its ports. Only authenticated users or devices are allowed to access the LAN through corresponding ports. Access control based on 802.1x prevents unauthenticated users or devices from accessing the network, which ensures transport network security.

To support access control based on 802.1x, Huawei base stations adopt unidirectional authentication based on Extensible Authentication Protocol-Transport Layer Security (EAP-TLS). The authentication is unidirectional because only the authentication server authenticates digital certificates of base stations. **Figure 3-1** shows the network topology for access control based on 802.1x.

Figure 3-1 Network topology for access control based on 802.1x



4 Access Control based on 802.1x

4.1 Principles

4.1.1 Working Principles

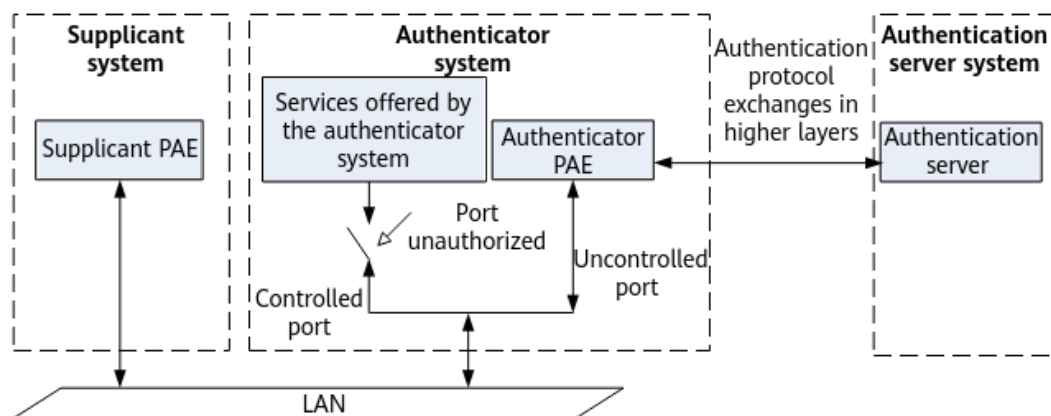
Access control based on 802.1x usually adopts the client-server architecture. The authentication access equipment forwards authentication packets from users or devices to the authentication server, and then the authentication server authenticates the identities of the users or devices. After successful authentication, the data flows of the users or devices can pass through the ports of the authentication access equipment.

Access control based on 802.1x involves the following components:

- Authentication client (embedded with a supplicant system): is a device to be authenticated, such as a base station, which initiates access control based on 802.1x. To support port-based access control, the authentication client needs to support the Extensible Authentication Protocol over LAN (EAPoL).
- Authentication access equipment (embedded with an authenticator system): is equipment (LAN switch, for example) that receives and forwards EAP authentication packets between the base station and authentication server at the Media Access Control (MAC) layer. The status of a controlled port (authorized or unauthorized) is determined by the authentication result of the authentication server.
- Authentication server (embedded with an authentication server system): authenticates clients. Commonly used servers include Remote Authentication Dial In User Service (RADIUS) servers and authentication, authorization and accounting (AAA) servers. The RADIUS and AAA servers have the same functions. This document uses the RADIUS server as an example.

Figure 4-1 shows the working principles of access control based on 802.1x.

Figure 4-1 Working principles of access control based on 802.1x



NOTE

Port access entity (PAE) is a port-related protocol entity that processes protocol packets during an authentication procedure.

A physical Ethernet port of the authentication access equipment can be used as both a controlled port and an uncontrolled port:

- **Controlled port:** A controlled port can be in the unauthorized or authorized state, depending on the authentication result at the authentication server.
 - A controlled port in the authorized state is bidirectionally connected and normal data flows can pass through the port.
 - A controlled port in the unauthorized state forbids any data to pass through.
- **Uncontrolled port:** An uncontrolled port is always in the bidirectional connectivity state and is mainly used to transmit EAPoL frames. This ensures that an authentication client can always transmit and receive authentication packets.

When initially visiting the transport network, a base station is not authenticated. The controlled port is in the unauthorized state. Only EAPoL packets can be sent to the authentication server through the uncontrolled port and other data cannot pass through the authentication access equipment. After the authentication server authenticates the base station and the authentication access equipment authorizes the controlled port, the controlled port changes into the authorized state and data from the base station can pass through the controlled port. This process ensures that only authenticated users and devices can visit the network.

Port-based access control can involve either a physical port (such as that identified by its MAC address) or a logical port (such as that of a VLAN). Huawei base stations support only port-based access control involving MAC addresses. In this case, an authentication message sent by a base station contains the MAC address of an Ethernet port that connects the base station to the transport network. After successful authentication, the authentication access equipment performs access control on data flows using this MAC address.

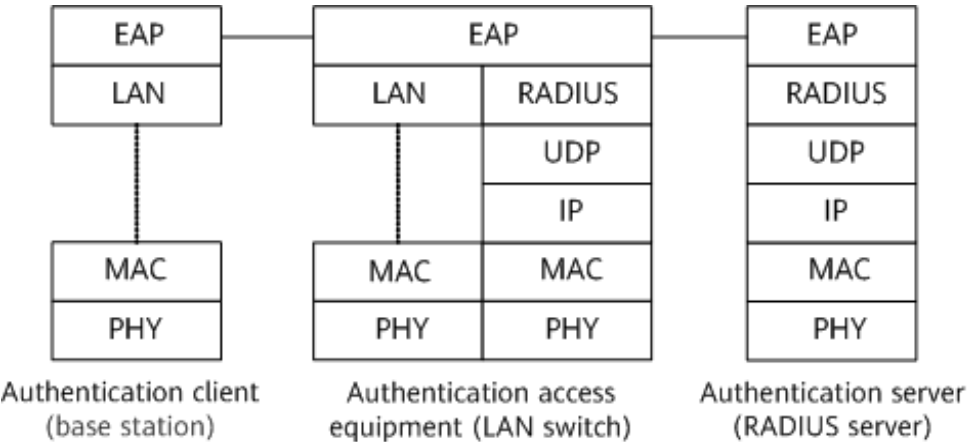
For details about access control based on 802.1x, see IEEE 802.1x-2004.

4.1.2 Protocol Stacks

To implement access control based on 802.1x, the authentication client and server exchange authentication messages using the EAP protocol. Between the authentication client and authentication access equipment, EAP data is encapsulated into EAPoL packets and then transmitted over the LAN. Between the authentication access equipment and authentication server, EAP data is encapsulated into EAP over RADIUS (EAPoR) packets and then transmitted using the RADIUS protocol.

Figure 4-2 shows the protocol stacks of three NEs.

Figure 4-2 Protocol stacks for access control based on 802.1x



Access control based on 802.1x uses the EAP protocol for identity authentication. The EAP protocol supports multiple authentication methods. Huawei base stations adopt unidirectional EAP-TLS authentication, indicating that only the authentication server authenticates base stations using digital certificates.

During access control based on 802.1x, a base station sends its digital certificate to the RADIUS server through an EAPoL packet. The RADIUS server authenticates the base station using the Huawei root certificate or the operator's root certificate. In this procedure, EAP-TLS unidirectional authentication is used, which is specified by **DOT1X.AM (5G gNodeB, LTE eNodeB)** (old model)/**DOT1XAUTH.AM (5G gNodeB, LTE eNodeB)** (new model).

When the base station functions as the EAP-TLS client in access control based on 802.1x, it supports TLS1.2 and the cipher suites listed in Table 4-1.

Table 4-1 Cipher suites supported by the base station functioning as the EAP-TLS client in access control based on 802.1x

SN	SSL Cipher Suite
1	ECDHE_ECDSA_AES_256_GCM_SHA384
2	ECDHE_RSA_AES_256_GCM_SHA384
3	ECDHE_ECDSA_AES_128_GCM_SHA256

SN	SSL Cipher Suite
4	ECDHE_RSA_AES_128_GCM_SHA256
5	DHE_RSA_AES_256_GCM_SHA384
6	DHE_RSA_AES_128_GCM_SHA256

NOTE

The preceding cipher suites are sorted based on the algorithm security strength and algorithm performance. A smaller sequence number (SN) indicates higher security of a cipher suite, which corresponds to a higher priority. It is recommended that these cipher suites be selected in a descending order of priority based on actual network requirements.

For details about the EAP protocol, see RFC 3748.

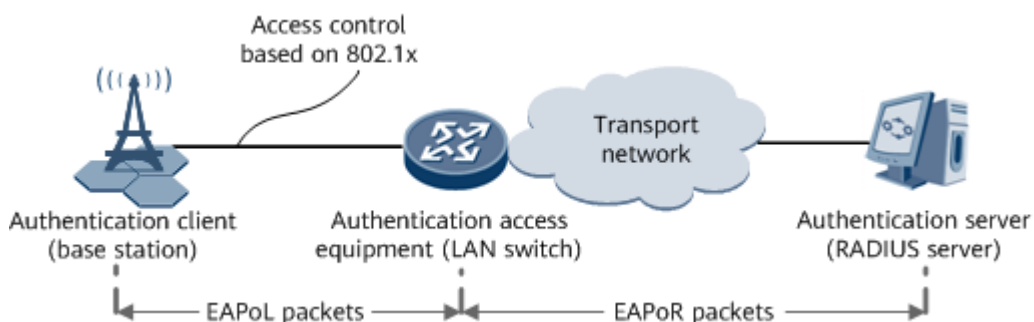
For details about the EAP-TLS protocol, see RFC 2716.

4.1.3 Application of Access Control Based on 802.1x

4.1.3.1 Typical Network Topology

To implement access control based on 802.1x on a network, an authentication access equipment (generally a LAN switch that is directly connected to a base station) supporting access control based on 802.1x and an authentication server are required. The authentication server needs to be preconfigured with the Huawei root certificate because Huawei base stations adopt unidirectional EAP-TLS authentication for access control based on 802.1x and have been preconfigured with Huawei initial certificates and the Huawei root certificate before delivery. [Figure 4-3](#) shows a typical network topology for access control based on 802.1x.

Figure 4-3 Typical network topology for access control based on 802.1x



Access control based on 802.1x can be activated on Ethernet ports using the **ACT DOT1X** (old model)/**ACT DOT1XAUTH** (new model) command and deactivated using the **DEA DOT1X** (old model)/**DEA DOT1XAUTH** (new model) command.

4.1.3.2 Auto-Discovery with Access Control based on 802.1x

4.1.3.2.1 Automatic Base Station Deployment by PnP

When access control based on 802.1x is activated on the network, a base station must be authenticated based on 802.1x before automatic deployment by plug and play (PnP). To ensure that the base station adapts to the network, Huawei base stations support deployment by PnP in the following scenarios:

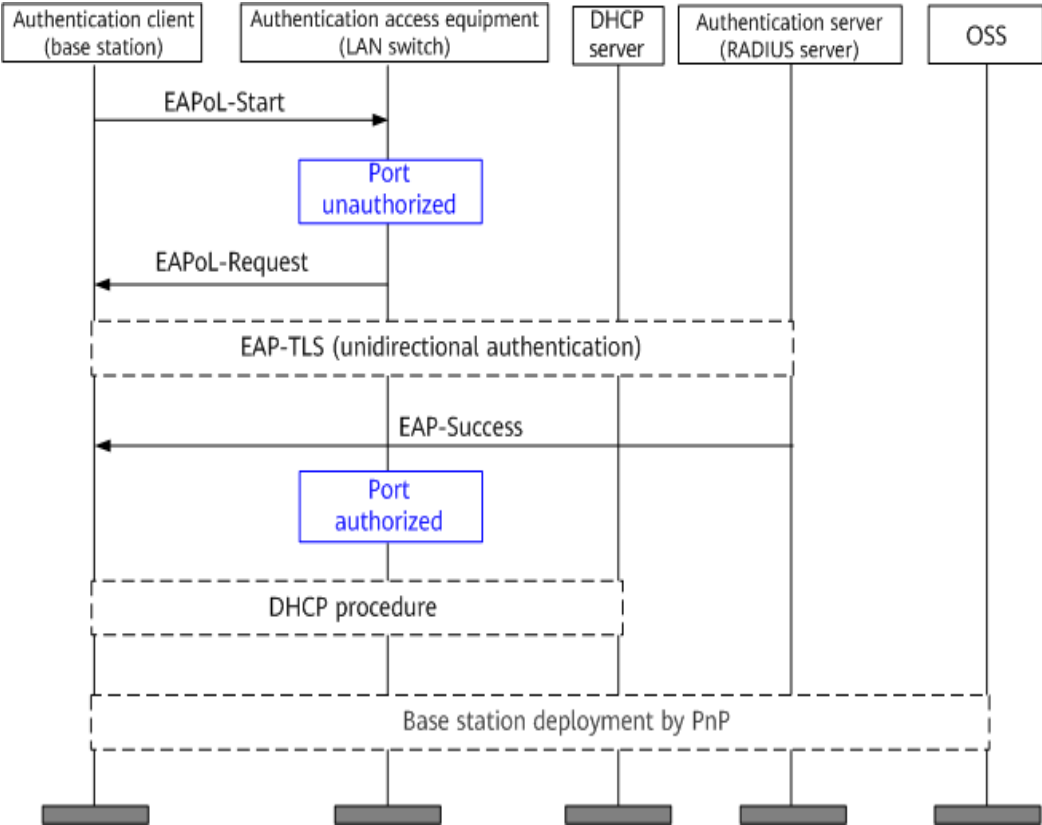
NOTE

During automatic base station deployment by PnP, base stations are authenticated for access control based on 802.1x using the preconfigured Huawei initial certificates.

Scenario 1

Figure 4-4 shows automatic base station deployment when the network supports access control based on 802.1x and access control based on 802.1x is activated on the Ethernet port that connects the base station to the transport network.

Figure 4-4 Automatic base station deployment (1)



The procedure is as follows:

1. After a base station is powered on, it sends an EAPoL-Start packet to the authentication access equipment to initiate access control based on 802.1x.
2. The base station, authentication access equipment, and authentication server perform access control based on 802.1x. The base station can initiate access control based on 802.1x on the same Ethernet port for a maximum of three times at an interval of 25 seconds.

3. If access control based on 802.1x is successful, the base station initiates a DHCP procedure. After the DHCP procedure is complete, the automatic base station deployment procedure starts.
4. If access control based on 802.1x fails, the base station initiates a DHCP procedure but fails to receive any response, leading to a DHCP procedure failure. The base station attempts to initiate access control based on 802.1x and a DHCP procedure on the next Ethernet port in round-robin mode.

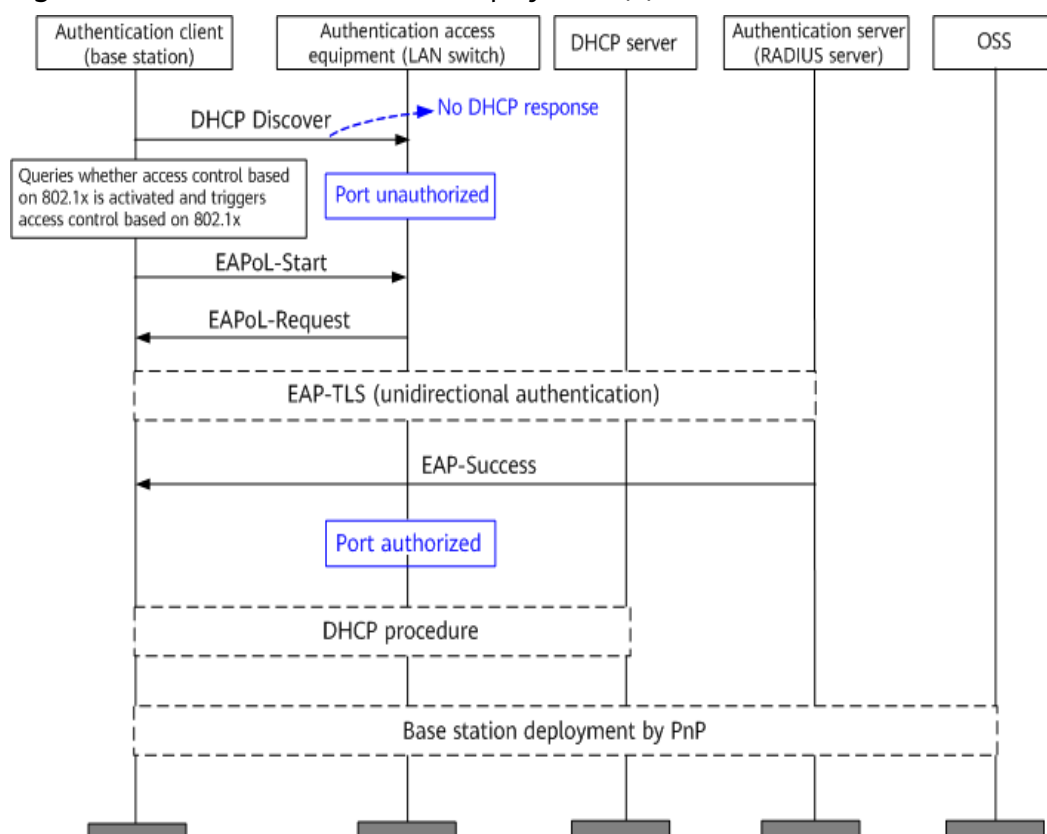
NOTE

During access control based on 802.1x, the EAPoL-Start packet is a multicast packet and its destination MAC address is 01-80-C2-00-00-03; other packets are unicast packets.

Scenario 2

Figure 4-5 shows automatic base station deployment when the network supports access control based on 802.1x but access control based on 802.1x is deactivated on the Ethernet port that connects the base station to the transport network.

Figure 4-5 Automatic base station deployment (2)



The procedure is as follows:

1. After a base station is powered on, it sends DHCP Discover packets to the authentication access equipment because access control based on 802.1x is deactivated on the Ethernet port that connects the base station to the transport network.
2. When sending DHCP Discover packets, the base station queries whether access control based on 802.1x is activated on the Ethernet port that connects

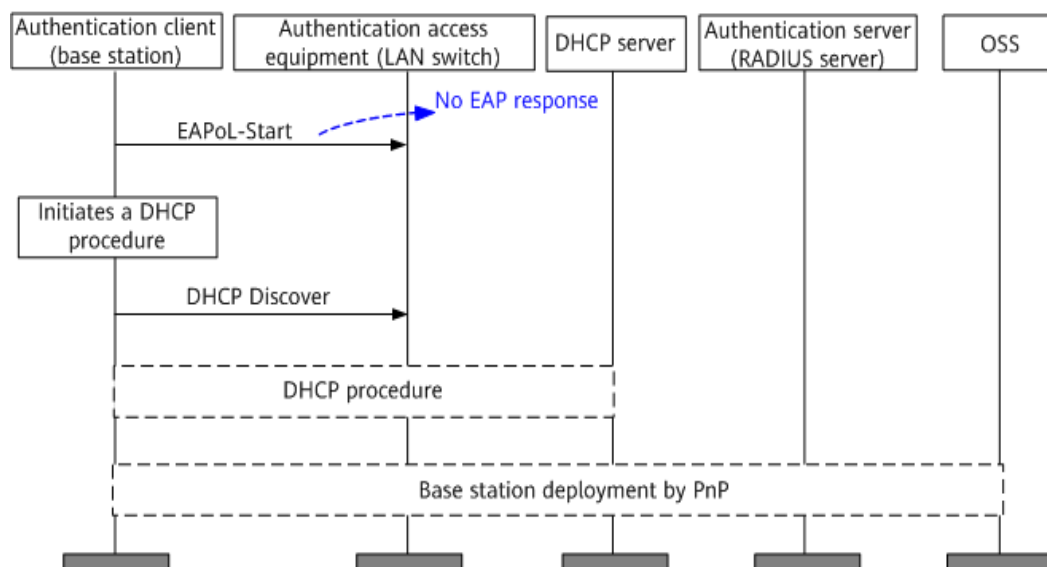
the base station to the transport network. If access control based on 802.1x is deactivated and authentication is not performed, the base station triggers access control based on 802.1x on this Ethernet port.

3. Because the controlled port of the authentication access equipment is in the unauthorized state, the base station cannot receive any response packets, leading to a DHCP procedure failure. The base station waits for the authentication result.
4. If access control based on 802.1x is successful, the base station retransmits DHCP Discover packets through the Ethernet port. After the DHCP procedure is complete, the automatic base station deployment procedure starts.

Scenario 3

Figure 4-6 shows automatic base station deployment when the network does not support access control based on 802.1x and access control based on 802.1x is activated on the Ethernet port that connects the base station to the transport network.

Figure 4-6 Automatic base station deployment (3)



The procedure is as follows:

1. After a base station is powered on, it initiates access control based on 802.1x. The base station sends an EAPoL-Start packet three consecutive times at an interval of 25 seconds but does not receive any response. Therefore, the base station determines that the network does not support access control based on 802.1x.
2. The base station sends DHCP Discover packets to the authentication access equipment.
3. After the DHCP procedure is complete, the automatic base station deployment procedure starts.

4.1.3.2.2 Application on Existing Base Stations

After a base station obtains the configuration file, it restarts. If the state of its Ethernet port changes from DOWN to UP and 802.1x-based access control is

activated on this Ethernet port, the base station initiates an 802.1x-based access control procedure. By default, 802.1x-based access control and SSL authentication use the same certificate:

- If the certificate used for SSL authentication in the configuration file is set to the operator-issued device certificate, the 802.1x-based access control procedure uses the operator-issued device certificate to authenticate the base station.
- If the certificate used for SSL authentication in the configuration file is set to the Huawei initial certificate, the 802.1x-based access control procedure uses the Huawei initial certificate to authenticate the base station.
- If the SSL authentication adopts the anonymous authentication mode, the 802.1x-based access control procedure uses the Huawei initial certificate by default to authenticate the base station.

4.2 Network Analysis

4.2.1 Benefits

Access control based on 802.1x prevents unauthenticated users or devices from accessing the network, which ensures transport network security.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.1.3 Function Impacts](#), [4.3.2.2.3 Function Impacts](#), [4.3.2.3.3 Function Impacts](#), [4.3.2.4.3 Function Impacts](#), [4.3.2.5.3 Function Impacts](#), and [4.3.2.6.3 Function Impacts](#).

When access control based on 802.1x is enabled, the time for base station deployment by PnP is extended by about 75s.

4.3 Requirements

4.3.1 Licenses

The license controlling access control based on 802.1x needs to be activated only for FDD, TDD, and NB-IoT eNodeBs. This function is not under license control on the GBTS, eGBTS, NodeB, and gNodeB.

RA T	Feature ID	Feature Name	Model	License Control Item Name	NE	Sales Unit
LTE FDD	LOFD-003015	Access Control based on 802.1x	LT1S000ACC00	Access Control based on 802.1x (FDD)	eNodeB	per eNodeB

RAT	Feature ID	Feature Name	Model	License Control Item Name	NE	Sales Unit
LTE CLoT	MLOFD-003015	Access Control based on 802.1x	ML1S000 ACC00	Access Control based on 802.1x (NB-IoT)	eNodeB	per eNodeB
LTE TD	TDLOFD-003015	Access Control based on 802.1x	LT1ST00 ACC00	Access Control based on 802.1x (TDD)	eNodeB	per eNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 LOFD-003015 Access Control based on 802.1x

4.3.2.1.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
LTE FDD	Public Key Infrastructure (PKI)	None	<i>PKI</i>

4.3.2.1.2 Mutually Exclusive Functions

None

4.3.2.1.3 Function Impacts

None

4.3.2.2 MLOFD-003015 Access Control based on 802.1x

4.3.2.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
NB-IoT	Public Key Infrastructure (PKI)	None	<i>PKI</i>

4.3.2.2.2 Mutually Exclusive Functions

None

4.3.2.2.3 Function Impacts

None

4.3.2.3 TDLOFD-003015 Access Control based on 802.1x

4.3.2.3.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
LTE TDD	Public Key Infrastructure (PKI)	None	<i>PKI</i>

4.3.2.3.2 Mutually Exclusive Functions

None

4.3.2.3.3 Function Impacts

None

4.3.2.4 FBFD-010023 Security Mechanism (Access Control based on 802.1x)

4.3.2.4.1 Prerequisite Functions

None

4.3.2.4.2 Mutually Exclusive Functions

None

4.3.2.4.3 Function Impacts

None

4.3.2.5 Access Control based on 802.1x on the GSM Side

4.3.2.5.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
GSM	BTS Supporting PKI	None	<i>PKI</i>
GSM	Abis over IP	None	<i>IPv4 Transmission</i>

4.3.2.5.2 Mutually Exclusive Functions

None

4.3.2.5.3 Function Impacts

None

4.3.2.6 Access Control based on 802.1x on the UMTS Side

4.3.2.6.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
UMTS	NodeB PKI Support	None	<i>PKI</i>
UMTS	IP Transmission Introduction on Iub Interface	None	<i>IPv4 Transmission</i>

4.3.2.6.2 Mutually Exclusive Functions

None

4.3.2.6.3 Function Impacts

None

4.3.3 Hardware

Base Station Models

RAT	Base Station Model
GSM	3900 and 5900 series base stations
UMTS	3900 and 5900 series base stations - DBS3900 LampSite and DBS5900 LampSite

RAT	Base Station Model
LTE	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
NR	3900 and 5900 series base stations. 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910. Only NR TDD is supported.

Boards

NE Type	Board Configuration	Type of Port Connecting to the Transport Network
eGBTS	UMPT/UMDU providing a transmission port	Ethernet port
eGBTS	UMPT+UTRPc, with the UTRPc providing a transmission port	Ethernet port
NodeB	UMPT/UMDU/MDUC providing a transmission port	Ethernet port
NodeB	UMPT+UTRPc, with the UTRPc providing a transmission port	Ethernet port
eNodeB	UMPT/UMDU providing a transmission port	Ethernet port
gNodeB	UMPT providing a transmission port	Ethernet port
USU	UEFU providing a transmission port NOTE The USU3910 uses the UEFU to provide a transmission port.	Ethernet port
Multimode base station	UMPT/UMDU/MDUC providing a transmission port NOTE The MDUC supports only GSM and UMTS dual-mode.	Ethernet port
Multimode base station	UMPT+UTRPc, with the UTRPc providing a transmission port	Ethernet port

RF Modules

No requirements

4.3.4 Networking

- An authentication server has been deployed in the network.
- The authentication server supports the EAP protocol defined in RFC 3748 and supports EAP-TLS authentication.
- An operator-issued device certificate must be used for authentication in the operation phase. The operator's root certificate must be preconfigured on the authentication server.
- The authentication access equipment supports access control based on 802.1x and EAP packet processing.
- The authentication access equipment supports port-based access control based on the MAC address.

4.3.5 Others

None

4.4 Operation and Maintenance

4.4.1 When to Use

If the operator's transport network is an open network, the devices in the transport network are vulnerable to unauthorized access and malicious attacks. In this case, it is recommended that access control based on 802.1x be activated to authenticate the users or devices that attempt to access the transport network. This prevents unauthenticated users and devices from accessing the network and ensures transport network security.

Access control based on 802.1x uses Huawei-issued device certificate to authenticate base stations. Therefore, PKI also needs to be activated.

4.4.2 Data Configuration

Huawei base stations support only unidirectional EAP-TLS authentication and port-based access control based on the MAC address. Therefore, before you activate access control based on 802.1x, check whether the authentication server supports unidirectional EAP-TLS authentication and whether the authentication access equipment supports port-based access control based on the MAC address. In the customer operation phase, configure the digital certificate issued by the operator CA system on an NE to ensure that the operator-issued device certificate is used for 802.1x authentication. The PKI system needs to be deployed on the operator network. For details, see *PKI*.

4.4.2.1 Data Preparation

NOTE

"-" in this section indicates that there is no special requirement for setting the parameters. Set the parameters according to network plans.

Table 4-2 lists the data to be prepared in the **DOT1X** MO before you activate access control based on 802.1x when the **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **OLD**.

Table 4-2 Data to be prepared for activating access control based on 802.1x (old model)

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	DOT1X.CN (5G gNodeB, LTE eNodeB)	-
Subrack No.	DOT1X.SRN (5G gNodeB, LTE eNodeB)	-
Slot No.	DOT1X.SN (5G gNodeB, LTE eNodeB)	-
Subboard Type	DOT1X.SBT (5G gNodeB, LTE eNodeB)	-
Port No.	DOT1X.PN (5G gNodeB, LTE eNodeB)	-
Authentic Method	DOT1X.AM (5G gNodeB, LTE eNodeB)	This parameter indicates the authentication method used in access control based on 802.1x. This parameter can be set to support EAP-TLS authentication.

Table 4-3 lists the data to be prepared in the **DOT1XAUTH** MO before you activate access control based on 802.1x when the **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **NEW**.

Table 4-3 Data to be prepared for activating access control based on 802.1x (new model)

Parameter Name	Parameter ID	Setting Notes
802.1x Authentication ID	DOT1XAUTH.DOT1XAUT HID (5G gNodeB, LTE eNodeB)	-
Port Type	DOT1XAUTH.PT (5G gNodeB, LTE eNodeB)	-
Port ID	DOT1XAUTH.PORTID (5G gNodeB, LTE eNodeB)	-

Parameter Name	Parameter ID	Setting Notes
Authentic Method	DOT1XAUTH.AM (5G gNodeB, LTE eNodeB)	This parameter indicates the authentication method used in access control based on 802.1x. This parameter can be set to support EAP-TLS authentication.

NOTE

- When deploying this feature on a multimode base station, activate the feature only on the Ethernet port that connects the base station to the transport network. The data preparation and initial configuration of the multimode base station for feature deployment are the same as those of a single-mode base station.
- When a base station is working normally, the certificate used in access control based on 802.1x is the same as that used by SSL authentication. For details about how to configure the certificate for SSL authentication, see the "Data Configuration" section in *SSL*. If no certificate is configured for SSL authentication, access control based on 802.1x uses Huawei-issued certificates by default.

4.4.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Run the **ACT DOT1X** (old model)/**ACT DOT1XAUTH** (new model) command to activate access control based on 802.1x on the Ethernet port that connects the base station to the transport network.

The following is an MML command example when the **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **OLD**:

```
//Activating access control based on 802.1x on the Ethernet port that connects the NodeB/eNodeB/gNodeB/eGBTS to the transport network  
ACT DOT1X: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PN=0, AM=EAP-TLS;
```

The following is an MML command example when the **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **NEW**:

```
//Activating access control based on 802.1x on the Ethernet port that connects the NodeB/eNodeB/gNodeB/eGBTS to the transport network  
ACT DOT1XAUTH: DOT1XAUTHID=0, PT=ETH, PORTID=0, AM=EAP-TLS;
```

Deactivation Command Examples

Run the **DEA DOT1X** (old model)/**DEA DOT1XAUTH** (new model) command to deactivate access control based on 802.1x on the Ethernet port that connects the base station to the transport network.

The following is an MML command example when the **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **OLD**:

```
//Deactivating access control based on 802.1x
DEA DOT1X: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PN=0;
```

The following is an MML command example when the **GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** parameter is set to **NEW**:

```
//Deactivating access control based on 802.1x
DEA DOT1XAUTH: DOT1XAUTHID=0;
```

4.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.3 Activation Verification

Run the **DSP DOT1X** (old model)/**DSP DOT1XAUTH** (new model) command to query whether access control based on 802.1x is activated on the Ethernet port that connects the base station to the transport network. Check the value of the **Authentic State** parameter in the command output. If the value of this parameter is **Authenticate Succeed**, access control based on 802.1x has been activated on the port.

The following figure shows an example of the command output:

Figure 4-7 DSP DOT1X command output

```
DSP DOT1X: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PN=0;%%
RETCODE = 0 Operation succeeded.
Display 802.1x
-----
Cabinet No. = 0
Subrack No. = 0
Slot No. = 7
Subboard Type = Base Board
Port No. = 0
Authentic Method = EAP-TLS authentic method
Authentic State = Authenticate Succeed
Authentic Succeed Number = 1
Fail Number = 0
Fail Reason = 0
Send EAP Packet Number = 7
Receive EAP Packet Number = 7
Abnormal Packet Number = 0
(Number of results = 1)
```

4.4.4 Network Monitoring

None

5 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

1. IETF RFC 3748, "Extensible Authentication Protocol (EAP)"
2. IETF RFC 2716, "PPP EAP TLS Authentication Protocol"
3. IEEE Std 802.1x-2004, "Port-Based Network Access Control"
4. *PKI* for SingleRAN
5. *SSL* for SingleRAN
6. *SRAN Networking and Evolution Overview* for SingleRAN
7. *IPv4 Transmission* for SingleRAN
8. *License Control Item Lists (FDD), License Control Item Lists (CIoT), and License Control Item Lists (TDD)* in *eRAN Feature Documentation*